

S1 True. Let r be the distance of every particle in the rigid body to the rotational axis, then we have $I_{xx} = \sum m r^2$.

$$\begin{aligned} \text{The kinetic energy } T &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} m (r \omega)^2 \\ &= \frac{1}{2} m r^2 \omega^2 \\ &= \frac{1}{2} I_{xx} \omega^2 \end{aligned}$$

$\therefore$ It is true

S2 False. Since the wheel is moving, the friction should not be $\mu_s N$ which is for static. Actually, when the wheel is moving along a horizontal rough surface, frictional force should be $\mu_k N$ where μ_k is the kinetic friction coefficient between the wheel and rough surface.

S3 False. If the collision happened between two ball is perfectly elastic where $e=1$, then the total kinetic energy of the system will be conserved, it will not be reduced. So it may be reduced but it will not when the collision is perfectly elastic.

S4 (a) $m \ddot{x} + kx = 0$

the types of vibration is Simple Harmonic Motion.

(b) $m \ddot{x} + kx = k \delta_m \sin \omega_f t$

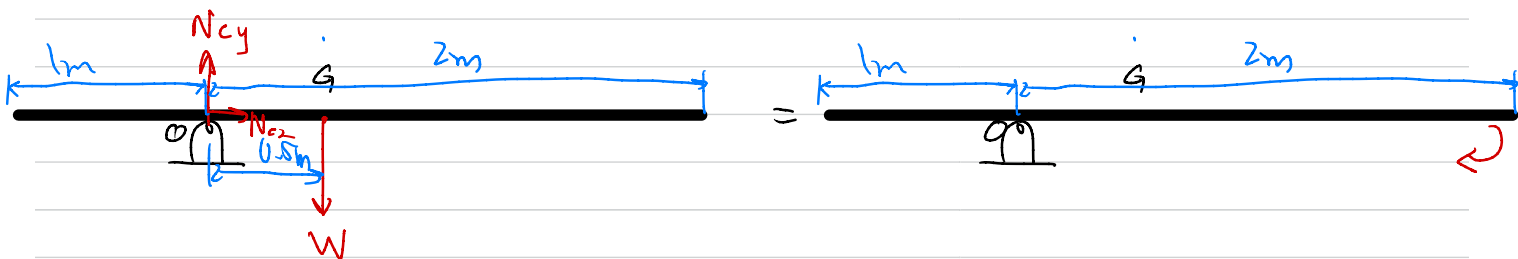
the types of vibration is Forced Vibration where ω_f is forced frequency.

(c) $m \ddot{x} + c \dot{x} + kx = 0$

the types of vibration is Damped Free Vibration.

Problem 1

(a) FBD



(b)

From the POD, $I_0 = \frac{1}{3} m l^2 - m r^2$
 $= \frac{1}{3} 20 \times 3^2 - 20 \times 1^2$
 $= 40$

By $\sum M_0 = \sum M_{eff}$
 $W(0.5) = I \alpha$

$$20 \times 9.81 \times 0.5 = 40 \alpha$$

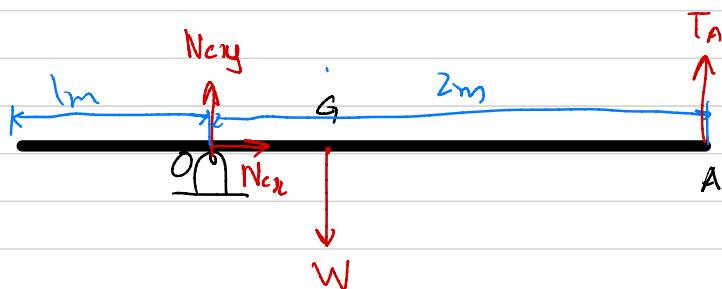
$$\alpha = 2.4525 \text{ rad/s}^2$$

$$\therefore a_A = r \alpha$$

$$= 3 \times 2.4525$$

$$= 7.3575 \text{ m/s}^2$$

(c)



when the cable is attached, the rod is in equilibrium

$$+\circlearrowleft \sum M_0 = 0 \Rightarrow W(0.5) - T_A(2.5) = 0$$

$$T_A = \frac{20 \times 9.81 \times 0.5}{2}$$

$$= 49.05 \text{ N}$$

Consider along x-axis, $\sum F_x = 0$, $N_{cx} = 0$

Consider along y-axis, $\sum F_y = 0$, $N_{cy} + T_A = W$

$$N_{cy} = W - T_A$$

$$= 20 \times 9.81 - 49.05$$

$$= 147.15 \text{ N}$$

$\therefore$ The reaction at pin support is $147.15 \text{ N} \uparrow$

Problem 2

By the conservation of energy,

$$E_1 = E_2$$

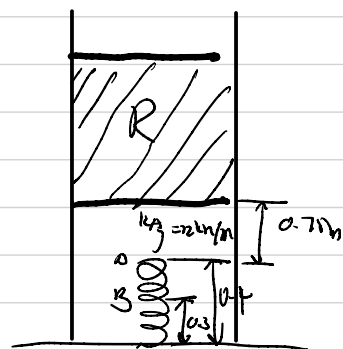
$$mgh = \frac{1}{2} k_A (0.1 + x)^2 + \frac{1}{2} k_B x^2$$

$$100 \times 9.81 \times 0.15 = \frac{1}{2} \times 12 \times 10^3 \times (0.1 + x)^2 + \frac{1}{2} \times 15 \times 10^3 \times x^2$$

$$135.75 = 6000 (0.01 + 0.2x + x^2) + 7500x^2$$

$$135.75 = 60 + 1200x + 6000x^2 + 7500x^2$$

$$13500x^2 + 1200x - 675.75 = 0$$

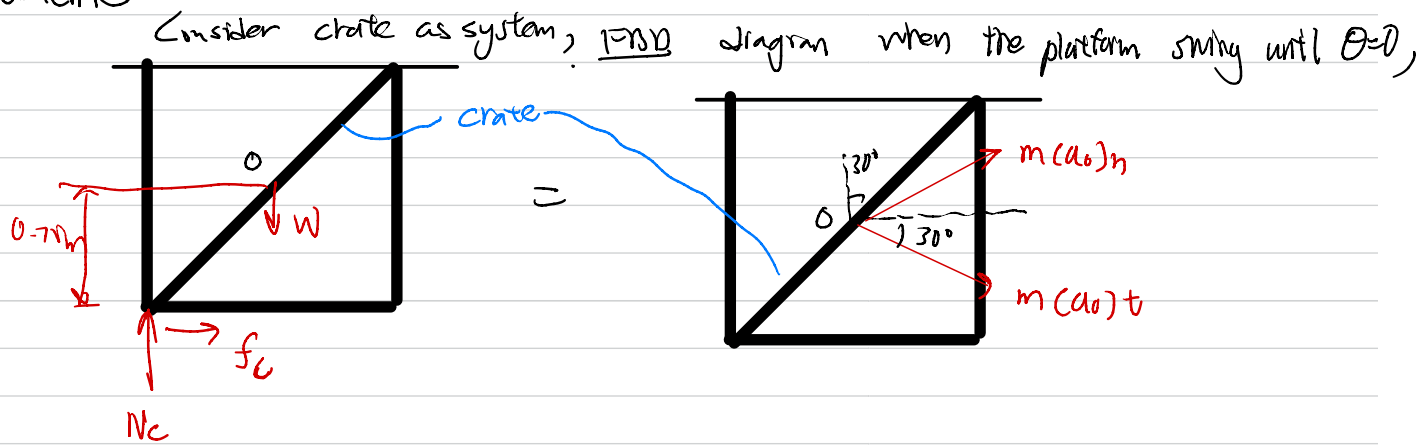


By solving the eqn, we have $x = 0.18 \text{ m}$

$$\therefore \text{max displacement of A} : x_{Amax} = x + 0.1$$

$$= 0.28 \text{ m}$$

Problem 3



From the FBD, $(a_0)_n = \omega^2 r$
 $= 4(1)^2$
 $= 4 \text{ m/s}^2$

$(a_0)_t = r\alpha$
 $= 4(0.5)$
 $= 2 \text{ m/s}^2$

Consider along $\rightarrow x$ -direction, $\sum F_x = m(a_0)_x$
 $\Rightarrow f_c = m(a_0)_n \sin 30^\circ + m(a_0)_t \cos 30^\circ$
 $f_c = 50 \times 4 \sin 30^\circ + 50 \times 2 \cos 30^\circ$
 $= 186.60 \text{ N}$

Along $\uparrow y$ -direction, $\sum F_y = m(a_0)_y$
 $N_c - W = m(a_0)_n \cos 30^\circ + m(a_0)_t \sin 30^\circ$
 $N_c = 50 \times 9.81 + 50 \times 4 \cos 30^\circ + 50 \times 2 \sin 30^\circ$
 $= 613.7 \text{ (N)}$

For checking, ① $f_{c, \max} = \mu_s N$
 $= 0.5 (613.7)$
 $= 306.9 \text{ N} \quad (> 186.6 \text{ N})$

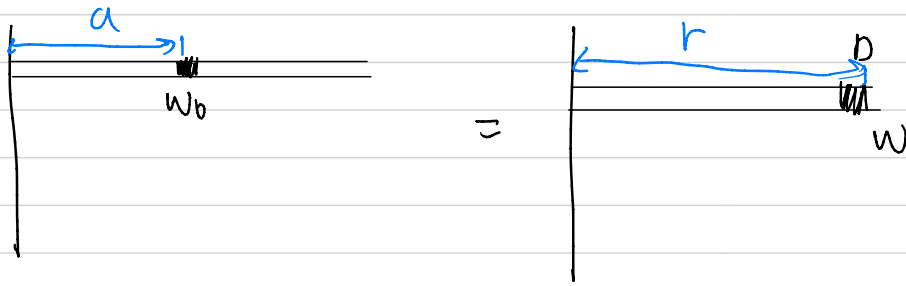
② By $\sum M_o = \sum (M_k)_o \Rightarrow N_c(x) - f_c(0.75) = 0$
 $613.7x - (186.6)(0.75) = 0$
 $x = 0.228 \text{ m} \quad (< 0.25 \text{ m})$

$\therefore$ By ① & ②, crate is rest when $f_c = 186.6 \text{ N}$

$\therefore f_c = 186.6 \text{ N}$ is the answer

Problem

4



(1) by the conservation of angular momentum,

$$H_1 = H_2$$

$$m_{\text{ball}} a \omega_0 = m_{\text{ball}} r \omega$$

$$\omega = \frac{a}{r} \omega_0$$

(2)

Since the ball is doing circular motion,

$$F = m r \omega^2$$

$$= m_{\text{ball}} r \left(\frac{a}{r} \omega_0 \right)^2$$

$$= \frac{m_{\text{ball}} a^2 \omega_0^2}{r}$$